

Name: _____

SOLVING LINEAR EQUATIONS, (TWO-STEP EQUATIONS)

Date: _____

LO: I can solve two-step linear equations by undoing operations in reverse order.

What is a two-step equation?

A **two-step equation** needs **TWO** inverse operations, e.g. $3x + 4 = 19$ subtract 4, then divide by 3.

Order matters: always undo +/- FIRST, then $\times/\div$. This is the reverse of BIDMAS.

Strategy

●	$3x + 4 = 19$	-4 then $\div 3$
●	$x = 5$ (check: $15+4=19$)	always check
●	Undo +/- first	reverse BIDMAS
●	$3x + 4 = 19$ $\div 3$ first	wrong order
●	Forget to do both sides	must do BOTH

Key idea

Undo the LAST operation first. For $3x + 4$: the last operation done to x was $+4$, so undo $+4$ first.

$$\begin{array}{c} \textcircled{3x + 4} = 19 \quad \textcircled{3x} = 15 \quad \textcircled{x = 5} \\ -4 \text{ then } \div 3 \end{array}$$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - $+/\times$

Solve $3x + 4 = 19$

Subtract 4 from both sides

$$3x = 15$$

Divide both sides by 3 $x = 5$

$$x = 5$$

Check: $3(5) + 4 = 15 + 4 = 19$

Example 2 - $-/\times$

Solve $5x - 7 = 18$

Add 7 to both sides

$$5x = 25$$

Divide by 5 $x = 5$

$$x = 5$$

Check: $5(5) - 7 = 18$

Example 3 - $+/\div$

Solve $x \div 3 + 2 = 6$

Subtract 2 from both sides

$$x \div 3 = 4$$

Multiply both sides by 3 $x = 12$

$$x = 12$$

Check: $12 \div 3 + 2 = 4 + 2 = 6$

Example 4 - fractional answer

Solve $4x + 3 = 10$

Subtract 3 $4x = 7$

Divide by 4 $x = \frac{7}{4}$

Or $x = 1.75$

$$x = \frac{7}{4}$$

Some answers are fractions or decimals.

YOUR TURN – PRACTICE (deliberate practice)

1. +/× form

Subtract first, then divide.

- a) $2x + 3 = 11$
- b) $5x + 7 = 32$
- c) $4x + 8 = 32$
- d) $7x + 1 = 50$

2. -/÷ form

Add first, then divide.

- a) $3x - 5 = 16$
- b) $6x - 2 = 22$
- c) $4x - 9 = 19$
- d) $8x - 3 = 45$

3. Division form

x is divided. Undo the +/- first, then multiply.

- a) $x \div 2 + 3 = 8$
- b) $x \div 4 - 1 = 3$
- c) $x \div 5 + 2 = 7$
- d) $x \div 3 - 2 = 4$

4. Fractional answers

Don't be put off by fractions or decimals.

- a) $4x + 3 = 10$
- b) $3x - 2 = 8$
- c) $5x + 1 = 8$
- d) $6x - 5 = 7$

5. Mixed two-step

Choose the right order of operations.

- a) $9x + 7 = 70$
- b) $x \div 6 + 4 = 9$
- c) $12x - 5 = 31$
- d) $x \div 2 - 7 = -3$
- e) $11x + 9 = 86$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $3x + 4 = 19$ first divide by 3

☐

Correct answer: _____

Reason: _____

b) Subtract 4 first, THEN divide

☐

Correct answer: _____

Reason: _____

c) $5x - 7 = 18$ means $x = 5$ ☐

Correct answer: _____

Reason: _____

d) Undo BIDMAS in reverse order

☐

Correct answer: _____

Reason: _____

e) x can never be a fraction ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A taxi charges **£3 fixed fee** plus **£2 per mile**. A journey costs **£17**. Form an equation $2x + 3 = 17$ and solve to find the distance.

Expression: _____

Simplified: _____

b) I think of a number, multiply by 4, then subtract 9. The result is 23. Form a two-step equation and solve to find my number.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

